

Institute of Actuaries of India

Subject SP2 – Life Insurance Principles

June 2019 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i) The key factors that influence the amount of initial capital needed to write a contract are:

Relationship between Pricing and Reserving Basis –

The reserving basis is generally said to be stronger or more prudent than the pricing/premium basis.

This implies that the premium less initial expenses are lower than the reserves needed to be set up at the inception of the contract. This extra amount or margin hidden in the reserve is released over time as experience unfolds but results in capital strain at inception of the contract.

Thus if the company ends up calculating the reserves at the same basis used for premium calculation there will be no initial capital strain. However stronger the reserving basis compared to the pricing basis more initial capital would be needed to write the contract.

Contract Design –

Sometimes the contract design enables insurance companies to keep reserves low or match the income and outgo at inception

In other words lower the initial reserves or better matched the cash flow position at inception, lower is the initial capital requirement

For example consider a unit linked endowment assurance product, where a low percentage of premiums are allocated to the unit fund at contract inception via high premium allocation charges. This ensures that the unit liability is kept low and at the same time also ensures that most of early premiums are made available to meet the high initial expense outgo resulting in a lower capital strain and sometimes even a profit

Frequency of Premium Payment –

A lower premium modal frequency results in less premium being available at inception to cover the high initial expenses. This is compensated to an extent by lower reserves as we take credit for the future modal premiums receivable in our reserve calculation.

However in case the policyholder dies or withdraws prior to contract termination this high initial expense may not be covered and thus the allowance for future premiums in the reserve calculation does not provide full protection as compared to say a single premium contract where all premiums are paid at the point of sale.

Additional Solvency Capital Requirement –

In most countries the regulatory authority would require insurance companies to hold extra assets in the form of supervisory capital over and above assets that are held to back the liabilities.

This explicit capital requirement forms an extra layer of protection for the policyholders in case of experience being worse than expected. However this extra supervisory or regulatory capital results in a higher initial capital strain as insurers need to demonstrate that they have sufficient assets to back the required solvency capital. For companies that do not sufficient level of extra assets or capital, an extra amount of capital in the shareholders fund needs to be injected, for them to continue to write more new business.

The manner of computation of solvency capital could be risk based or a factor-based model, but the need to hold additional solvency capital at inception increases the initial capital strain.

Level of Initial Expenses –

The level of initial expenses has a direct influence on the amount of initial capital needed to write a contract. The higher the level of initial expenses the lower would be the asset share. This also in theory can be compensated by holding lower reserves at inception which takes credit for the future expense loadings in our future premium.

However in practice, the initial expenses are often higher than that loaded for. This results in initial capital strain to the extent of the extra initial expenses which are not covered by the loadings in our future premiums in the reserve calculation.

Also there is generally a restriction on reserves to be positive or have a minimum value of zero. This results in initial capital strain when the asset share is negative due to high initial expenses as the reserves cannot be allowed to fall beyond the minimum allowable level of zero.

[10]

ii) The items which contribute to the determination of asset share of a policy apart from the ones listed are:

- a. Cost of any death benefits
- b. Cost of any other benefits under the product like waiver of premium etc
- c. Surrender profits or losses
- d. Charges for any options or guarantees
- e. Share of any non participating business profits or losses written under the participating fund
- f. Tax
- g. Transfer of shareholders share of profit
- h. Cost of capital required to support the contract
- i. Transfer to free assets to support a smoothing policy

Additionally, the items already listed can be further extended as

Investment income – into investment income, realised capital gain/losses, unrealised capital gain/losses

Commission – into initial commission, renewal commission

Expenses – into initial, renewal and termination expenses and further into premium related, sum assured related, per policy related, investment management related etc.

[4]

[14 Marks]

Solution 2:

i)

Policy data to price the option and Business volume

Since the product is new, the insurer may not have relevant data to estimate the option take up rate needed to price the option...

..It might be difficult to get this data from external sources as well..

This would make the insurer vulnerable to a different option take up rate than priced for.

The profit from this product will be highly sensitive to the option take up behaviour by the policyholder.

Business volume

The risk of loss is aggravated if option rates are not estimated correctly and the product becomes very popular and sells in large volumes.

Alternatively too little business due to the complexity of the product could lead to non-recovery of the overhead costs involved in managing the option.

Mortality basis

The mortality rate of the policyholders choosing the updated version of the product is expected to be higher than those who choose the base version of the product especially after the option take up date.

This is a risk if the premium basis fails to reflect the correct mortality rate experienced by the policyholder post the option date.

Using reinsurance may mitigate some of the mortality risk..

...but counterparty risk is created as the insurer is still liable to make claim payments even when the reinsurer fails to honour its commitments

Competition

The product may not sell if another insurer introduces this product in the market before our company.

The competition could also force the insurer to lower the premium rates to match that of the competitors...

...this could prove extremely risky as a new untested product is being launched without sufficient prudent margins in the premiums charged

Distribution Channel

The distributors may try to exploit any loopholes in the product design.

..this could lead to anti selection which would invalidate the mortality assumption used to price the product.

Alternatively there is a risk that some distribution channels like own sales force may not completely understand the complex product design and thus are not able to explain it to the clients

This could also lead to mis-selling which could damage the company's reputation.

Mix of business by size

There is expected to be some element of expense cross subsidy from bigger ticket size policies to smaller ticket size policies.

If the insurer sells higher proportion of lower ticket size policies the recovery of overheads could be an issue.

Policy administration system

The data management system of the company may not be capable of capturing option take up exercised by the policyholders who choose the updated version of the product.

This functionality needs to be developed and could lead to significant costs being incurred in system development which would then need to be passed on to the policyholders

There is a risk that product proves to be unpopular...

... this would make recovery of the development cost difficult.

Higher Capital Requirements

As the product category is new, the regulator may prescribe higher than normal margins in the reserves or enforce additional capital requirement to ensure that the claim payments to policyholders under all scenarios are protected.

This could lead to an ongoing higher capital requirement to support the product than estimated at the point of sale.

Underwriting

Poor underwriting practices or controls at policy inception leading to sub standard lives being accepted at normal rates.

The presence of an option to increase the SA by 100% after one year further aggravates the risk.

Expenses

The expense associated with administering the option could be higher than anticipated

If the policyholder does not continue the policy in the second year, the expenses associated with underwriting may not be recouped

Some other risks in the product design are deviation from the pricing basis for each of

- Withdrawal rate
- Investment return
- Expense Inflation etc

But all of these are unlikely to be very significant given the short-term nature of the product

Legal, regulatory risk

Since this product has only just been permitted, further changes in regulations could cause further development costs at a later stage or could make the product unprofitable

[13]

ii) Premium Payable by policyholders, who do not choose the option,

$$\begin{aligned}
 A &= 10000000 * 1.2 / 1000 * 1/1.06 + 10000000 * (1 - 1.2/1000) * (1.3/1000) * 1/1.06^2 \\
 &= 11320.75 + 11556.07 \\
 &= 22876.82
 \end{aligned}$$

Premium paid at the start of the second year at normal premium rates by the option takers for increasing the SA to 20000000,

$$\begin{aligned}
 B &= 10000000 * 1.3 / 1000 * 1/1.06 \\
 &= 12264.15
 \end{aligned}$$

Let us now draw two life tables based on actual mortality experienced for those who do not take the option and those who do take the option using a radix of 100000:

Table1

Base Table			
Age	lx	qx	dx
35	100000.00	0.001200	120.00000
36	74910.00	0.001300	97.38300
37	74812.62		

Lx at age 36 in the base table excludes the 25% of lives which would opt for the option

Table 2

Option Holders table			
Age	lx	qx	dx
35			
36	24970.00	0.001950	48.69150
37	24921.31		

25% of the people surviving to age 36 choose the option and the option holder's then experience 150% of the normal mortality rates

Using table 1 and table 2 the total EPV of the benefits is

$$10000000 * 0.001200 * 1/1.06 + 10000000 * 0.000977383 * 1/1.06^2 + 20000000 * 0.000486915 * 1/1.06^2$$

$$= 28654.86 \quad \dots\dots\dots (i)$$

Similarly, EPV of premiums payable is

$$A + 0.24970 * 1/1.06 * B$$

$$= 25765.84 \quad \dots\dots\dots (ii)$$

So the value of the additional option premium payable at inception is $= 28654.86 - 25765.84 = 2889.02$

[12]

[25 Marks]

Solution 3:

i) The value assigned to the mortality rate parameter should reflect the expected experience of the lives who will take out the term insurance contract being sold.

The expected experience would depend crucially on the following factors:

- Target market
- Distribution channel
- Expected change in experience from the last experience investigation to the point in time at which the assumptions would apply
- Underwriting controls applied

The final mortality rate used would be a combination of a base rate, mortality trend and possibly some margin applied to these rates

The base rate would generally be expressed as some percentage or adjustment to rates from a standard table if available.

Given that the company is venturing into the term insurance product for the first time, the company would not have in-house experience for this product.

A suitable alternative such as experience on a whole life contract may be used.

The data should relate to an appropriate number of years, such that the volume of data is adequate but excessive heterogeneity due to trends over time is not introduced.

Split the data into relevant homogeneous groups by age, gender, medical /non medical, smoker/non smoker, distribution channel, geographical area etc.

This should be subject to adequate data being retained within each cell.

If the insurer has insufficient data to produce reliable results, or has no appropriate data, use other sources of data such as industry level data or reinsurer's data to form an estimate.

Industry sources of data may be good for showing trends, since trends in the insurers own data may be put down to statistical variation.

Since the product is very popular in the country, relevant industry data should be available, and the reinsurer would also have experience of the product.

Similarly, population mortality statistics may be useful in picking up trends if it has been examined at regular intervals in the past.

However since we intend to determine mortality rates for a term assurance product the usefulness of such data may be limited due to difference in experience between population data and the insured population.

Also if the rates are to apply to a class of lives which are expected to have a different experience from that to which the analysed data relates, further adjustment may be necessary...

... this could arise due to changes in target market, distribution channel or underwriting standards in the future.

Since this is a new product and the company may not have relevant data, extra margins may be added to the mortality rates to reflect the risk of adverse experience than expected.

The extent of margins used generally depends upon

- The level of confidence in the base mortality assumption
- The financial significance of the mortality assumption
- The need to be competitive

It is also a common practice to allow for the margin in pricing via the risk element of the risk discount rate than the individual mortality rate parameter assumption. [9]

ii) Reserve value

The per policy reserves (and asset share) for regular pay term insurance plans tend to be low.

Therefore, even if surrender values were to be offered, they would be very small, and not viewed as valuable by the policyholder.

However, for single pay term insurance this might not be the case.

For regular pay term insurance, the policyholder often perceives the premium each year to be paying for the life cover in that year. Therefore the policyholder expectation for surrender value does not exist.

Whereas, for single pay product, the customer may expect part of the premium to be returned, pertaining to the outstanding period of cover

The other players in the market may be offering a surrender value for single pay, creating customer expectation [3]

iii) Methods of calculation of surrender values: (i) The retrospective method and (ii) The Prospective Method

The Retrospective method requires sufficient up to date information to calculate earned asset share value for each policy. From the asset share value, the surrender value is arrived.

However, this being a term insurance product, the insurer may not maintain the asset shares, hence may calculate a retrospective reserve using a formula.

Suitable parameters, which reflect the experience of the term insurance products in terms of mortality and expenses, cost of surrender, may be required to calculate the retrospective reserve for each duration of the policy.

The second method, the prospective method, estimates the value of future benefits and expenses, net of future premiums due.

Suitable parameters estimates, which reflect future expected investment returns, expenses, and mortality experience of the surrendering policyholders and the corresponding costs of surrender, may be required to calculate the prospective reserve for each duration of the policy, [6]

iv) Analysis of the suitability of methods:

Generally, the retrospective reserve value represents the earned asset share or an estimate as on the date of surrender. Hence, this provides the maximum that the insurer can pay without making a loss. At early durations care need to be taken that it should not look too unreasonable compared with the premiums paid.

As for term insurance products, retrospective method using asset share is not suitable to estimate that surrender value. Hence, insurers may use formula for arriving at the retrospective reserve method.

However, the retrospective method may produce possibly very low values at very short durations, depending on the relationship between premium, initial expenses and commission.

Thus, unless the policyholders understand the deduction for initial expenses well, their expectations may not be met.

The surrender values, generally, are expected to run into the maturity value. However, with respect to the surrender value for a term insurance policy, there being no maturity value, as duration in-force approaches the full term, the surrender value may have to be zero. At some point, there could be a possibility that the surrender value may be more favourable than continuing the policy, which may encourage lapse and re-entry.

But this method ignores the future benefits and expected future experience.

To that extent, the surrender value calculated may be significantly different from a realistic prospective value.

However, the insurer may not have sufficient historic information to determine suitable parameters gathered on all the existing policies to calculate the retrospective reserve for arriving at the surrender values at each duration. Hence, the insurer may not be in a position to use this method to calculate surrender value for existing policies.

With regard to the prospective method, if the insurer uses realistic basis, the resultant surrender value represents what the contract is worth to the company.

The surrender value may appear unreasonable at early durations to the policyholders, if the surrender value is very low or negative. The surrender value is dependent on the on the basis used

The prospective method is relatively easy when compared to the retrospective method as it does not require any knowledge of what has happened in the past. [10]

v) The data from the app may not be exhaustive, since only part of the population uses the app. Even if the data is available for all customers, there could be reliability issues.

Is the app being used by the life to be insured, or a family member, for example

Is the data accurate, and trustworthy?

It could open the company to counterparty risk as well. If the service provider shuts down, the underwriting data will suddenly not be available.

The data that may be made available from the health service providers may not be specific and structured and may pose risk in identifying relevant data for underwriting from the data that is shared.

The quality, completeness and the credibility of the data may not be known and to that extent relying only on the data that is captured by the health service provider may be a concern

Current underwriting assessment methodologies may not be suitable given the unstructured data
New underwriting methodologies may need to be developed and the staff need to be trained – may be exposed to the risk of incorrect method being developed or the staff may find it difficult to switch from the current method to the new method difficult.

There could be too many rejections or acceptance, based on the possibility of capturing more information and possible detailed analysis, virtually changing the overall product design parameters.

The possibility of anti-selection, by managing the data being captured data with the health service provider.

More data, more underwriting, harsh decisions, possibly incurring more resource time spent and costs, not in similar lines what competitors do – bad publicity or reduced business or reduced sales
Risk of not getting reinsurance at reasonable rates or availability of reinsurance, if this is perceived as a risky strategy by the reinsurer

The impact of the proposed change would be more severe as the insurer is making the change for a term assurance product as the risk of anti selection and loss to the insurer is greater for such a product

Possibility of increase in the premium rates, due to perceived higher involvement of resources, higher costs, development of new methodologies, and non-availability o increase in reinsurance rates
May have to review claims underwriting – The insurer may want to review the claims underwriting to depending on how the new underwriting methodology is developed like if relaxed underwriting, then strengthen claims underwriting or make changes in the product design

It might be better solution to supplement the data from the app with the current underwriting practices, and to phase out the proposal form if the decision of both methods starts to coincide. [10]

[38 Marks]

Solution 4:

i) Assuming that there are no restrictions in the current product structure, the following may help the insurer to manage the guarantee:

The number of unit linked funds offered may be limited to one or two to limit the scope of switching from one fund to another

The funds that allow for risky investments may be avoided

The guarantee may be made applicable only on funds which only invest in bond markets. If the policyholder switches from the bond fund to any other fund within the product, the underlying guarantee will be lost during the remaining policy term.

The insurer needs to keep complete flexibility in switching the asset category from one asset to another within the fund without any specific limits on various asset categories, if a combined is offered, to enable to take advantage of asset movements as well as to enable to take preventive steps in case of adverse movements in asset categories;

The guarantee should not be made applicable, if the policyholder surrenders the policy; [5]

ii) Since the cost of investment guarantee depends upon future investment returns, a stochastic investment model should be the most appropriate to use.

Since a stochastic model is complex there can be a number of checks on the model.

It is important to check the model logic and code...

All the important assumptions should be validated....

...the most important assumption in this model is the expected fund growth rate and volatility and should be verified independently for correctness and reasonableness.

Check if the probability distribution of the investment return assumption used in the model is appropriate and independently verifiable with any other available data.

If model points are used, the results of the model should be tested for model point error. This would involve running the model at different model points and checking that the results do not differ significantly.

Check whether the model points are based on actual in force policy data.

This might involve scaling up the model points and checking if they provide a close approximation to the actual unit value and number of policies at the date of valuation.

Check the cash flows of a single run from the stochastic model by building a spreadsheet model using the same parameters as used for that run.

Check the stochastic model using deterministic investment return assumption...

...for example a high investment growth rate should result in zero cost while at the same time lowering the investment growth assumption should gradually lead to an increase in the cost of the guarantee.

This could be used to find out the break even investment return which could then be checked for reasonableness.

Check if the model gives sensible result if the frequency or time period of projection is altered from say 1 year to 6 months.

Check whether the numbers of randomly generated investment scenarios are enough and that the results are not very sensitive to the number of simulations used.

Check whether there is a dynamic link between parameters for example future unit growth rates and withdrawal rates.

Check whether varying the other parameters of the model like mortality, withdrawals and paid up produces sensible results...

...these parameters would generally be deterministic and reflect the expected value of these parameters in the future.

Check if the method of calculating the cost of guarantee is in line with any regulatory guidance

Check if the cost of guarantee is reasonable by comparing it with the market price of a suitable derivative.

[8]

[13 Marks]

Solution 5:

i) The principle of investment states the fundamental approach towards balancing risk and return.

It can be expressed as:

- To minimise risk, a company should select investments that are appropriate to the nature, term and currency of the liabilities
- The investments should also be selected to maximise the overall return on the assets, where overall return includes both investment income and capital gains
- The extent to which the appropriate investments referred to above may be departed from in order to maximise the overall return will depend, amongst other things, on the extent of the company's free assets and the company's appetite for risk.

These principles can also be expressed as

- The company should invest so as to maximise the overall return on the assets, subject to the risk being within the financial resources available to it. [3]

ii)

Advantages

A minimum investment of 50% in government and state securities ensures a fairly low default risk as government bonds are the most secure form of investment. These investments also generally have a deep and liquid market.

An investment of not more than 35% in AA rated and above corporate bonds allows for low default risk. This protects policyholders money invested in the company.

Also this allows policyholders to benefit from a higher return from corporate bonds as yields on corporate bonds are expected to be higher than government bonds of the same duration.

Investment in equities is expected to allow policyholders to receive a real return on their investments..

.. maximum cap on equity investment protects policyholders from fluctuations in equity markets.

Also restriction in investments to quoted stocks ensures that the investment is marketable and liquid in nature.

Since there is no overseas investment, currency risk or mismatching of assets and liabilities by currency is zero.

Disadvantages

The norms seem to be fairly prudent in terms of the overall investment strategy.

This would impact the overall expected returns on participating business even though the returns to policyholders on non-participating and non-linked products remain unaffected.

The investment norms may lead to lower expected returns made available to policyholders as

- At least 50% of the investments are in govt bonds which offer the lowest expected return amongst investment asset classes

- No investment in property is allowed. This reduces the investment return that the company may earn...
... this also reduces the scope to gain from any diversification benefits that could be achieved
- unquoted equities even though risky may provide a higher expected return which is lost out as no investment in this category is allowed
- Sometimes investment in foreign assets may provide significant diversification benefits as well as the opportunity to benefit from favourable exchange rate movements. This opportunity is also lost as no investment in foreign currency is allowed.
- restricting the equity component to 15% may reduce the ability of the company to earn real returns...
.... this can lead to policyholder payouts not moving in line with inflation

[7]

[10 Marks]
